

Cyclotomic filters and a $\mathbb{Z}_{37}$ orbit-matrix attack on the order-668 Hadamard problem

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June 2026

Abstract

A Hadamard matrix of order $668 = 4 \cdot 167$ is the smallest order for which existence is currently unknown. We study two of the equivalent objects whose construction would settle the question: Legendre pairs of length $\ell = 333$, and the strongly regular (conference) graph $\text{srg}(333, 166, 82, 83)$, equivalent to a symmetric conference matrix $C(334)$. We make three contributions. First, we give a clean account of the power-spectral conditions for a Legendre pair of length $333 = 9 \cdot 37$ and prove that the natural *square-divisor* refinement of the Kotsireas–Koutschan cyclotomic filter (evaluating the spectrum at a primitive 9th root of unity) provides *no* additional pruning beyond the cube-root (mod 3) filter; the refinement lives in the cubic field $\mathbb{Q}(\zeta_9)^+$ but is implied by the cube-root condition. Second, assuming a fixed-point-free automorphism of order 37 (which the eigenvalue-field obstruction *permits*, unlike the group-developed case), we reduce $\text{srg}(333, 166, 82, 83)$ to a 9×9 array of circulants over $\mathbb{Z}_{37}$, with the strong-regularity identity equivalent to a per-character condition $M_t^2 + M_t - 83I = 0$ on 9×9 Hermitian matrices over $\mathbb{Z}[\zeta_{37}]$. Third, we prove by SAT that *no* cyclotomically-structured (Paley-type) such graph exists for 7 of the 8 proper nontrivial multiplier subgroups: completely for every subgroup of order ≥ 3 (for the six orders ≥ 4 independently of any orbit matrix, and for order 3 by exhausting the 960 compatible orbit matrices); for the last (order 2) we verify nonexistence over 634 enumerated skeletons. We also give large-scale parallel-tempering evidence (a search validated to reconstruct $\text{srg}(25, 12, 5, 6)$) that none admits an unstructured lift either. These results narrow the search for $H(668)$ and we release validated open-source machinery for the reduction.

1 Introduction

A *Hadamard matrix* of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ and $HH^\top = nI$. The Hadamard conjecture asserts that such a matrix exists for every $n \equiv 0 \pmod{4}$. The smallest order for which existence is open is $n = 668 = 4 \cdot 167$ [1, 2]. Several classical objects are each equivalent to $H(668)$; we work with two.

Legendre pairs. For $x \in \{\pm 1\}^\ell$ let $\text{PAF}_x(k) = \sum_{i \in \mathbb{Z}_\ell} x_i x_{i+k}$ be the periodic autocorrelation. A *Legendre pair* (LP) of length ℓ is a pair $u, v \in \{\pm 1\}^\ell$ with $\text{PAF}_u(k) + \text{PAF}_v(k) = -2$ for all $k \not\equiv 0$. An LP of length ℓ yields a Hadamard matrix of order $2\ell + 2$ via the two-circulant-core construction [3]. For $H(668)$ one needs $\ell = 333 = 9 \cdot 37$; LP(333) is open [4, 5].

Conference graphs. A symmetric conference matrix C of order $m = 334$ ($C = C^\top$, zero diagonal, ± 1 off-diagonal, $C^2 = (m - 1)I$) yields $H(668)$ via $H = \begin{bmatrix} C+I & C-I \\ C-I & -C-I \end{bmatrix}$, and $C(334)$ is equivalent to a regular two-graph on 334 points, whose descendant is the strongly regular graph $\Gamma = \text{srg}(333, 166, 82, 83)$ [7, 2]. The parameter set $(333, 166, 82, 83)$ is listed as open in Brouwer's tables, and no catalogued conference-matrix construction (Paley; Mathon's $q^2(q+2)+1$ family [8]) reaches order 334.

Prior obstructions. It is known that no *group-developed* (Cayley) conference graph on 333 vertices exists: its nontrivial eigenvalues are $(-1 \pm \sqrt{333})/2$ with $\sqrt{333} = 3\sqrt{37} \in \mathbb{Q}(\sqrt{37})$ of conductor 37, while every group of order $333 = 3^2 \cdot 37$ has commutator subgroup inside the Sylow-37 subgroup and hence admits a character of order 3; a Cayley eigenvalue $\chi(S)$ of order-3 character lies in $\mathbb{Z}[\zeta_3] \not\subseteq \mathbb{Q}(\sqrt{37})$, a contradiction (Lemma 1). Likewise the relevant prime 2 is not self-conjugate modulo 167 ($\text{ord}_{167}(2) = 83$), so the Turyn–Schmidt field-descent bounds do not forbid $H(668)$; the barrier is constructive, not arithmetic.

Contributions. We (i) settle the question of whether a square-divisor cyclotomic filter strengthens the Kotsireas–Koutschan filter for LP(333) (Section 2); (ii) give a $\mathbb{Z}_{37}$ orbit-matrix reduction of Γ permitted by the eigenvalue-field obstruction, distinct from the group-developed case (Section 3); and (iii) report a complete computational attack on the reduced problem with rigorous (SAT) and heuristic (parallel-tempering) negative results (Section 4).

2 Cyclotomic filters for LP(333)

Write $\text{PSD}_x(s) = |\sum_j x_j \omega^{js}|^2$, $\omega = e^{2\pi i/\ell}$. Summing the LP condition gives the flat-spectrum form

$$\text{PSD}_u(s) + \text{PSD}_v(s) = 2\ell + 2 \quad (s \not\equiv 0), \quad \text{PSD}_u(0) + \text{PSD}_v(0) = 2, \quad (1)$$

so the row sums are ± 1 ; for $\ell = 333$ the constant is $2\ell + 2 = 668$.

For $d \mid \ell$ the d -compression $c_r^{(d)} = \sum_{j \equiv r \pmod{d}} x_j$ satisfies $\text{PAF}_{c^{(d)}}(k) = \sum_{i \equiv k \pmod{d}} \text{PAF}_x(i)$ [6], and $\text{PSD}_x(\ell/d)$ depends only on $c^{(d)}$.

Cube root (mod 3). At $s = \ell/3$, write $\gamma = c^{(3)}(u)$, with γ_r odd (a sum of $\ell/3 = 111$ values ± 1). Then $\text{PSD}_u(\ell/3) = |\gamma_0 + \gamma_1\zeta_3 + \gamma_2\zeta_3^2|^2 = \gamma_0^2 + \gamma_1^2 + \gamma_2^2 - \gamma_0\gamma_1 - \gamma_1\gamma_2 - \gamma_2\gamma_0$, the Eisenstein norm. Since the γ_r are odd and $1 + \zeta_3 + \zeta_3^2 = 0$, this value is divisible by 4: $\text{PSD}_u(\ell/3) = 4(x^2 + xy + y^2)$ with $x = (\gamma_0 - \gamma_1)/2$, $y = (\gamma_1 - \gamma_2)/2 \in \mathbb{Z}$. With (1) this gives the necessary condition

$$\frac{\ell+1}{2} = 167 = L_u + L_v, \quad L_u, L_v \text{ Loeschian (values of } x^2 + xy + y^2\text{)}. \quad (2)$$

A short enumeration recovers exactly the Kotsireas–Koutschan spectrum [4]: 16 admissible values of $\text{PSD}_u(\ell/3)$, all $\equiv 4 \pmod{12}$, forming 8 complementary Loeschian pairs summing to 167.

Ninth root (mod 9): the square-divisor refinement. Since $9 \mid 333$, one may also evaluate at a primitive 9th root $s = \ell/9$. With $c = c^{(9)}(u)$ (entries odd), the same parity argument gives $\text{PSD}_u(\ell/9) = 4|E(\zeta_9)|^2$ for $E \in \mathbb{Z}[\zeta_9]$, a totally non-negative element of the cubic field $K = \mathbb{Q}(\zeta_9)^+ = \mathbb{Q}(\cos \frac{2\pi}{9})$, and (1) reads $167 = N_u + N_v$ in $\mathcal{O}_K$ with N_u, N_v totally non-negative. This is a genuine cubic-field condition, and no published filter uses a prime-power (rather than prime) frequency. One might hope it strengthens (2). It does not.

Proposition 1 (Square-divisor filter adds no pruning). *For every length $\ell \in \{27, 45, 63\}$ (i.e. $\ell = 9m$ with $m \in \{3, 5, 7\}$), every mod 3-admissible state (γ, γ') (a pair of odd triples with row sums ± 1 and $\sum \gamma_r^2 + \sum \gamma_r'^2 = 4m + 2$) is the 3-compression of a valid length-9 compressed Legendre pair (c, c') . Hence the ninth-root condition is implied by the cube-root condition and provides no additional pruning of the spectrum.*

Verification. For each ℓ we enumerate all length-9 integer pairs (c, c') with odd entries in $[-m, m]$, row sums ± 1 , and $\text{PAF}_c(k) + \text{PAF}_{c'}(k) = -2m$ ($k \neq 0$), $= 2\ell - 2m + 2$ ($k = 0$); we form the set of 3-compressions $(c^{(3)}, c'^{(3)})$ and check it equals the full set of mod 3-admissible states. The counts are: $\ell = 27$: 250,128 pairs, all 288 states realized; $\ell = 45$: 1,163,808 pairs, all 288 realized; $\ell = 63$: 9,385,956 pairs, all 756 realized. In every case the number of mod 3-admissible states that fail to lift is 0. $\square$

Remark 1. The case $\ell = 45$ has $(\ell + 1)/2 = 23$, prime and $\equiv 2 \pmod{3}$ (the same arithmetic type as $167 = (333 + 1)/2$), so the negative result is tested on a length representative of 333, not merely on small toys. Structurally, the 9-compression space is enormous and is a weak filter; the cube-root norm already captures all spectral information available at the 3-part. We conclude that effort on LP(333) should not be directed at square-divisor spectral filters.

3 A $\mathbb{Z}_{37}$ orbit-matrix reduction of $\text{srg}(333, 166, 82, 83)$

Lemma 1. *No group of order 333 admits a $(333, 166, 82, 83)$ partial difference set; equivalently there is no group-developed conference graph on 333 vertices.*

Sketch. As in the introduction: a 1-dimensional character of order 3 would force a nontrivial eigenvalue $(-1 \pm 3\sqrt{37})/2 \in \mathbb{Z}[\zeta_3]$, impossible since $\mathbb{Q}(\sqrt{37}) \not\subseteq \mathbb{Q}(\zeta_3)$. $\square$

Lemma 1 rules out the *regular* automorphism group; it does not rule out individual automorphisms. We therefore consider a fixed-point-free automorphism σ of order 37. Since $37 \mid 333$ ($333 = 9 \cdot 37$), such σ partitions the vertices into 9 orbits of size 37; we identify each orbit with $\mathbb{Z}_{37}$ so that σ acts as $+1$. The adjacency matrix becomes a 9×9 array of circulants: writing $D[i][j] \subseteq \mathbb{Z}_{37}$ for the connection set of block (i, j) , $A_{(i,a),(j,b)} = 1 \iff (b - a) \bmod 37 \in D[i][j]$, with $D[j][i] = -D[i][j]$ (symmetry) and $0 \notin D[i][i] = -D[i][i]$ (no loops). Crucially the eigenvalue-field obstruction of Lemma 1 does *not* apply: $\sqrt{37} \in \mathbb{Q}(\zeta_{37})$ (the quadratic Gauss sum), so order 37 is permitted. The reduction of a strongly regular graph to circulant blocks under a semiregular automorphism is the standard orbit-matrix method [10, 11]; our contribution is its application to this open parameter set, the resulting per-character condition, and the (non)existence results below.

Proposition 2 (Character reduction). *Define, for $t \in \mathbb{Z}_{37}$, the 9×9 matrix $M_t[i][j] = \sum_{d \in D[i][j]} \zeta_{37}^{td}$. Then A is $\text{srg}(333, 166, 82, 83)$ (equivalently $A^2 + A - 83I = 83J$) if and only if*

$$(t = 0) \quad R := M_0 \text{ satisfies } R^2 + R - 83I_9 = 3071J_9, \quad (t \neq 0) \quad M_t^2 + M_t - 83I_9 = 0,$$

where each M_t ($t \neq 0$) is Hermitian over $\mathbb{Z}[\zeta_{37}]$ with spectrum in $\{(-1 \pm \sqrt{333})/2\}$.

The integer matrix $R = (|D[i][j]|)$, the *orbit matrix*, is symmetric with row sums 166, even diagonal, and (from $R^2 + R - 83I = 3071J$) eigenvalues 166 (once) and $(-1 \pm \sqrt{333})/2$ (four times each); in particular $\text{tr } R = 162$ and a Cauchy–Schwarz argument confines the diagonal entries to $\{10, 12, \dots, 26\}$. We verified Proposition 2 by reconstructing the Paley conference graphs $P(9), P(25), P(49)$ in block-circulant form, confirming agreement of the direct and per-character tests.

4 Computational attack and results

Orbit-matrix census. A backtracking search with knapsack-tight pruning enumerated 634 distinct admissible orbit matrices R , over which the order-2 sweep of Theorem 1(d) ranges; a deeper search (per-diagonal cap 4×10^9 nodes) confirms at least 1678 admissible skeletons, with the true number larger still (a few diagonals have search trees exceeding 4×10^9 nodes and were not exhausted). The skeleton thus exists in abundance, which is exactly why the complete, enumeration-free arguments of Theorem 1(a) matter: they sidestep the (apparently intractable) full enumeration. The full enumeration *does* become tractable once a coset congruence is imposed: restricting to the order-3-compatible alphabet (diagonal in $\{12, 18, 24\}$, off-diagonal $\equiv 0, 1 \pmod{3}$) collapses the search to exactly 960 orbit matrices, enumerated to completion, the finite set that powers Theorem 1(b). The remaining obstruction is the *lift* to connection sets satisfying Proposition 2. A representative skeleton (row sums 166, $\text{tr} = 162$, spectrum $\{166, r^4, s^4\}$) is

$$R = \begin{pmatrix} 10 & 19 & 19 & 19 & 19 & 20 & 20 & 20 & 20 \\ 19 & 14 & 22 & 22 & 20 & 15 & 15 & 18 & 21 \\ 19 & 22 & 14 & 22 & 20 & 15 & 18 & 21 & 15 \\ 19 & 22 & 22 & 14 & 20 & 15 & 21 & 15 & 18 \\ 19 & 20 & 20 & 20 & 18 & 24 & 15 & 15 & 15 \\ 20 & 15 & 15 & 15 & 24 & 20 & 19 & 19 & 19 \\ 20 & 15 & 18 & 21 & 15 & 19 & 24 & 17 & 17 \\ 20 & 18 & 21 & 15 & 15 & 19 & 17 & 24 & 17 \\ 20 & 21 & 15 & 18 & 15 & 19 & 17 & 17 & 24 \end{pmatrix}, \quad R^2 + R - 83I_9 = 3071 J_9.$$

Exact lift (SAT). The lift is equivalent to the convolution system, for all $i, j \in \mathbb{Z}_9$, $g \in \mathbb{Z}_{37}$,

$$\sum_{i'} (D[i][i'] * D[i'][j])(g) = \mu + (k - \mu)[i=j, g=0] + (\lambda - \mu)[g \in D[i][j]],$$

which we encode in a Boolean SAT/CP model [9] (membership variables, cardinality, symmetry, and a gauge-fixing that removes the 37^8 per-orbit base-point symmetry). The encoding was validated by reconstructing $P(9)$ and $P(25)$.

Call a $\mathbb{Z}_{37}$ -symmetric $\text{srg}(333, 166, 82, 83)$ *H-cyclotomic* if every connection set is a union of cosets of a multiplier subgroup $H \leq \mathbb{Z}_{37}^*$ (the Paley-type structure). The next theorem rules these out for 7 of the 8 proper nontrivial subgroups completely (every subgroup of order ≥ 3), and for the remaining one (order 2) over our enumerated skeletons.

Theorem 1 (No cyclotomically-structured lift). *No H-cyclotomic $\text{srg}(333, 166, 82, 83)$ exists in the following cases.*

- (a) (Complete, via the orbit matrix; $|H| \in \{4, 6, 9, 12, 18, 36\}$.) An *H-cyclotomic* graph forces its 9×9 orbit matrix to have *H-coset-compatible* cardinalities: off-diagonal $R[i][j] \equiv 0$ or $1 \pmod{|H|}$ (the residue records whether $0 \in D[i][j]$), and diagonal $R[i][i] \equiv 0 \pmod{|H|}$ if $-1 \in H$ else $\pmod{2|H|}$. No integer matrix meets these together with the orbit-matrix identity $R^2 + R - 83I = 3071J$ (SAT-certified UNSAT); hence no such graph exists, for any orbit matrix.
- (b) (Complete, via exhaustive lift; $|H| = 3$.) Here $H = \{1, 10, 26\}$ is not closed under negation, forcing $R[i][i] \equiv 0 \pmod{6}$ and $R[i][j] \equiv 0, 1 \pmod{3}$. Unlike the cases in (a), such orbit matrices do exist: a complete backtracking enumeration (no node cap, 1.99×10^{10} nodes) finds exactly 960, all with diagonal $(12, 12, 18, 18, 18, 18, 18, 24, 24)$. For every one of the 960, the orbit-matrix-fixed order-3-invariant lift is UNSAT; hence no order-3-invariant graph exists.
- (c) (Complete, independent confirmation; $|H| \in \{12, 18, 36\}$.) The stronger orbit-matrix-free system (every block *H*-invariant, the graph 166-regular, R unconstrained) is directly UNSAT.

- (d) (*Over enumerated skeletons; $|H| = 2$.) For each of the 634 enumerated orbit-matrix skeletons, the order-2-invariant lift is UNSAT. (Order 2 imposes no congruence on the orbit matrix: every R is coset-compatible, so neither the argument of (a) nor the finite enumeration of (b) applies, and the result is a sample rather than a complete proof.)*)

Thus no Paley-type $\mathbb{Z}_{37}$ -symmetric $\text{srg}(333, 166, 82, 83)$ exists for *any* multiplier subgroup of order ≥ 3 , and none of order 2 over the enumerated skeletons. The SAT encodings were validated by reconstructing the Paley single circulants $P(13), P(17)$ under their quadratic-residue multiplier, and the orbit-matrix-free encoding by reconstructing $P(9), P(25), P(49)$.

Unstructured lift (parallel tempering). For the general (unstructured) lift we ran a batched GPU parallel-tempering search minimizing $\sum_{t \neq 0} \|M_t^2 + M_t - 83I\|_F^2$, which is 0 iff the lift succeeds. The search is validated: on the $\text{srg}(25, 12, 5, 6)$ skeleton it drives the objective to 0 (a verified lift) within 18 steps. On $\text{srg}(333, 166, 82, 83)$, across ≈ 450 of the 634 skeletons with 8192 chains and replica exchange per skeleton ($\approx 3.7 \times 10^6$ chains total), the objective plateaued at $\approx 3.9 \times 10^5$ with no lift found.

The order-2 residue: a sharper ceiling. The single case Theorem 1 leaves as a sample is $|H| = 2$ (negation-invariant blocks), where the orbit-matrix methods are powerless because order 2 imposes no congruence. It is worth recording how far the structure can in fact be pushed, and why it then resists. Since -1 is a quadratic residue mod 37, negation-invariance of every block makes each character matrix M_t a *real* symmetric 9×9 matrix with eigenvalues in $\{\theta, \tau\}$, $\theta, \tau = (-1 \pm 3\sqrt{37})/2$; write a_t for the θ -multiplicity of M_t . The Galois group $\mathbb{Z}_{37}^*/\{\pm 1\}$ permutes the M_t and acts on $\sqrt{37}$ through the Legendre symbol, forcing $a_t = a$ on the quadratic residues and $9 - a$ on the non-residues for a single integer a .

Proposition 3 (Order-2 multiplicity reduction). *For a negation-invariant $\mathbb{Z}_{37}$ -symmetric $\text{srg}(333, 166, 82, 83)$, the common residue-character θ -multiplicity satisfies $a \in \{4, 5\}$.*

Proof sketch. Expanding $\text{tr } M_1 = a\theta + (9 - a)\tau$ in the integral basis $\eta_g = \zeta_{37}^g + \zeta_{37}^{-g}$ via the quadratic Gauss sum $\sqrt{37} = \sum_{g=1}^{18} (\frac{g}{37}) \eta_g$ identifies the diagonal pair-counts $N_g := \#\{i : \{g, -g\} \subseteq D[i][i]\}$ as $N_g = 3a - 9$ on the nine residue pairs and $N_g = 18 - 3a$ on the nine non-residue pairs. Non-negativity $0 \leq N_g \leq 9$ gives $a \in \{3, 4, 5, 6\}$. For $a \in \{3, 6\}$ the counts are extremal $\{0, 9\}$, forcing every diagonal block to equal $Q^\mp$ (all non-residues, resp. all residues) and hence the orbit matrix to have constant diagonal $(18, \dots, 18)$; an exhaustive backtracking search, complete under the S_9 orbit-relabeling symmetry (1.9×10^8 nodes, corroborated to 9×10^{10} nodes by an independent reduction), finds *no* admissible orbit matrix with constant diagonal 18. Hence $a \in \{4, 5\}$. The N_g identity was validated on the negation-invariant graphs $P(13)$ and the 25-orbit Paley graph of $\text{GF}(5^3)$. $\square$

That the cases $a \in \{4, 5\}$ then resist all of our tools is structural, not accidental. In the group ring $\mathbb{Z}[\mathbb{Z}_{37}]$ (generator X) the strong-regularity identity reads $B^2 + B - 83I_9 = 83TE_9$, where $B = (B_{ij})$ has the block indicators as entries, E_9 is all-ones, and $T = \sum_g X^g$ is supported on the *principal* character. Every character-weighted (Gauss-/Jacobi-sum) functional therefore collapses to a plain cardinality already fixed by the orbit matrix; and since $83 \equiv 9 \pmod{37}$ has odd order 9 with $-1 \notin \langle 83 \rangle$, no self-conjugacy/multiplier obstruction applies. Moreover negation invariance makes $\varphi : (i, g) \mapsto (i, -g)$ a graph automorphism commuting with A , and as A^2 is affine in $\{I, A, J\}$, every word in $\{A, \varphi\}$ reduces to an affine combination of $\{I, A, J, \varphi, \varphi A\}$, whose traces are the fixed values $\{333, 0, 333, 9, 162\}$. Consequently *no* trace-linear invariant (spectral, character-theoretic,

involution-based, or association-scheme/Krein) separates a hypothetical solution from the orbit-matrix data: any obstruction must be genuinely non-linear in the $\{0,1\}$ block indicators, i.e. it resides in the lift feasibility itself, which is the order-668 problem. This is why the order-2 case is the true residue, and why we leave it open.

5 Discussion

We distinguish what is proven from what is evidence. *Proven:* Proposition 1 (square-divisor filters do not help LP(333)); Proposition 2 (the reduction); Theorem 1 (no cyclotomically-structured $\mathbb{Z}_{37}$ lift, complete for every multiplier subgroup of order ≥ 3); Proposition 3 (the remaining order-2 case is confined to residue-multiplicity $a \in \{4, 5\}$, and provably resists every trace-linear obstruction). *Evidence:* the parallel-tempering plateau is a strong indication, from a search demonstrably able to find lifts when they exist, that no unstructured $\mathbb{Z}_{37}$ -symmetric lift exists either, though it is not a proof. Taken together with Lemma 1, this makes a $\mathbb{Z}_{37}$ -symmetric $\text{srg}(333, 166, 82, 83)$ unlikely, and focuses any future construction of $H(668)$ on (a) genuinely asymmetric conference graphs, (b) automorphisms of other orders (for which the orbit-matrix block size grows), or (c) the Legendre-pair / cocyclic routes [12], where Section 2 clarifies which spectral tools do and do not help.

Reproducibility

All code (the $\mathbb{Z}_{37}$ reduction and verifier, the orbit-matrix enumerator and its order-3-compatible restriction, the exact SAT lift, the order-3 exhaustive lift sweep and the orbit-matrix-free cyclotomic check, the GPU batched parallel-tempering search, and the conference-to-Hadamard emitter), together with its validation (reconstruction of the Paley graphs $P(9), P(25), P(49)$, the conference-to-Hadamard chain on $H(12), H(28), H(36)$, and recovery of a genuine lift on the $\text{srg}(25, 12, 5, 6)$ skeleton) is openly available at <https://github.com/maceoCK/h668-z37> and archived on Zenodo (DOI to be assigned on release).

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